§ 3 two-Level

1. SG Experiment

$H = -\vec{S}\cdot\vec{B}$, $F_z = S_z \partial_z B_z$

Sequential SG

① z → $z+$ / $z-$ (blocked) → $z+$

② z → $z+$ / $z-$ (blocked) → $x+$ / $x-$

③ z → $z+$ / $z-$ (blocked) → x → $x+$ / $x-$ (blocked) → $z+$ / $z-$

2. Pauli matrix

$|z+\rangle = \begin{pmatrix}1\\0\end{pmatrix}$, $|z-\rangle = \begin{pmatrix}0\\1\end{pmatrix}$ $\quad \sigma_z|z\pm\rangle = \pm|z\pm\rangle \Rightarrow \sigma_z = \begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$|x+\rangle = \begin{pmatrix}1\\1\end{pmatrix}$, $|x-\rangle = \begin{pmatrix}1\\-1\end{pmatrix}$ $\quad \sigma_x|x\pm\rangle = \pm|x\pm\rangle \Rightarrow \sigma_x = \begin{pmatrix}0&1\\1&0\end{pmatrix}$

$|y+\rangle = \begin{pmatrix}1\\i\end{pmatrix}$, $|y-\rangle = \begin{pmatrix}1\\-i\end{pmatrix}$ $\quad \sigma_y|y\pm\rangle = \pm|y\pm\rangle \Rightarrow \sigma_y = \begin{pmatrix}0&-i\\i&0\end{pmatrix}$

$[\sigma_i,\sigma_j] = 2i\varepsilon_{ijk}\sigma_k$, $\vec{\sigma}^2 = 3$

$\{1,\sigma_x,\sigma_y,\sigma_z\}$ as basis

H : $H = H^\dagger = \begin{pmatrix}a&b\\b^*&c\end{pmatrix}$

空间内积 : $\langle A,B\rangle = \mathrm{Tr}(A^\dagger B)$

$$\begin{cases}\mathrm{Tr}(\sigma_\mu\sigma_\nu) = 2\delta_{\mu\nu}\\ \sum_m \sigma^{\uparrow}_{ab}\sigma^{\uparrow}_{cd} = 2\delta_{ad}\delta_{bc}\end{cases}$$

展开 : $H = h_0\sigma_0 + h_1\sigma_x + h_2\sigma_y + h_3\sigma_z$

其中 $h_\mu = \frac{1}{2}\mathrm{Tr}(\sigma_\mu H)$

For any H :

$H' = H - h_0 I = \vec{h}\cdot\vec{\sigma} \sim -\vec{B}\cdot\vec{\sigma}$

⇒ Bloch Dynamics

3. Constant Magnetic Field.

$$H = -2\cdot\frac{e}{2m}\frac{\hbar}{2}\vec{\sigma}\cdot\vec{B} = -\frac{e\hbar}{2m}\vec{\sigma}\cdot\vec{B} \;\rightarrow -\vec{\sigma}\cdot\vec{B}$$

$\vec{B} \parallel \hat{n}$.

$$H = -B\vec{\sigma}\cdot\hat{n} = -B\sigma_n$$

$$= -B\left(\sin\theta\cos\phi\,\sigma_x + \sin\theta\sin\phi\,\sigma_y + \cos\theta\,\sigma_z\right)$$

$$\sigma_n|n\pm\rangle = \pm|n\pm\rangle$$

① Eigenstate & eigenvalue.

$$\begin{vmatrix} \cos\theta & \sin\theta(\cos\phi - i\sin\phi) \\ \sin\theta(\cos\phi + i\sin\phi) & -\cos\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & \sin\theta\, e^{-i\phi} \\ \sin\theta\, e^{i\phi} & -\cos\theta \end{vmatrix}$$

$$(\lambda-\cos\theta)(\lambda+\cos\theta) - \sin^2\theta = 0 \;\rightarrow\; \lambda = \pm 1$$

$$\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\phi} \\ \sin\theta\, e^{i\phi} & \cos\theta \end{pmatrix}\begin{pmatrix} a_+ \\ b_+ \end{pmatrix} = \begin{pmatrix} a_+ \\ b_+ \end{pmatrix}$$

$$\begin{cases} \cos\theta\, a_+ + \sin\theta\, e^{-i\phi} b_+ = a_+ \\ \sin\theta\, e^{i\phi} a_+ + \cos\theta\, b_+ = b_+ \end{cases} \rightarrow \frac{a_+}{b_+} = \frac{1-\cos\theta}{\sin\theta}e^{-i\phi} = \tan\frac{\theta}{2}e^{-i\phi}$$

$$\text{Norm} \rightarrow |a_+|^2 + |b_+|^2 = 1 \rightarrow \begin{cases} a_+ = \cos\frac{\theta}{2}e^{-i\phi} \\ b_+ = \sin\frac{\theta}{2} \end{cases}$$

$$|n+\rangle = \begin{pmatrix} \cos\frac{\theta}{2}e^{-i\phi} \\ \sin\frac{\theta}{2} \end{pmatrix},\quad |n-\rangle = \begin{pmatrix} \sin\frac{\theta}{2}e^{-i\phi} \\ -\cos\frac{\theta}{2} \end{pmatrix}$$

② Rotation Method

$$\hat{U}(\theta,\hat{n}) := \exp\left(-\frac{i\theta}{2}\vec{\sigma}\cdot\hat{n}\right)$$

$$\sigma_n = \hat{U}\sigma_z\hat{U}^\dagger,\quad \theta = \langle\vec{\sigma}\cdot\vec{\hat{z}}\rangle$$

$$\sigma_n|n\pm\rangle = \pm|n\pm\rangle$$

$$\hat{U}^\dagger\sigma_n\hat{U}\hat{U}^\dagger|n\pm\rangle = \pm\hat{U}^\dagger|n\pm\rangle \quad\Rightarrow\quad \sigma_z|z\pm\rangle = \pm|z\pm\rangle$$

$$|n\pm\rangle = \hat{U}|z\pm\rangle$$

$$|n+\rangle = U_z(\phi)U_y(\theta)|z+\rangle = \begin{vmatrix} \cos\frac{\theta}{2}e^{-i\phi} \\ \sin\frac{\theta}{2}e^{i\phi} \end{vmatrix}$$

$$\langle n\pm|\vec{\sigma}|n\pm\rangle = \pm\vec{B}$$

4. Dynamic Evolution

$|\psi(0)\rangle = |n,\pm\rangle \Rightarrow |\psi(t)\rangle = e^{-iE_z t/\hbar}|n,\pm\rangle$, $E_z = \mp B$

let $\vec{B} = B\hat{z}$

$$|\psi(0)\rangle = \begin{pmatrix} \cos\frac{\theta}{2} e^{-i\phi} \\ \sin\frac{\theta}{2} \end{pmatrix}, \quad |\psi(t)\rangle = \begin{pmatrix} e^{+iBt/\hbar}\cos\frac{\theta}{2} e^{-i\phi} \\ e^{-iBt/\hbar}\sin\frac{\theta}{2} \end{pmatrix}$$

$$\langle \hat{\sigma}_x \rangle = \langle\psi(t)|\hat{\sigma}_x|\psi(t)\rangle$$

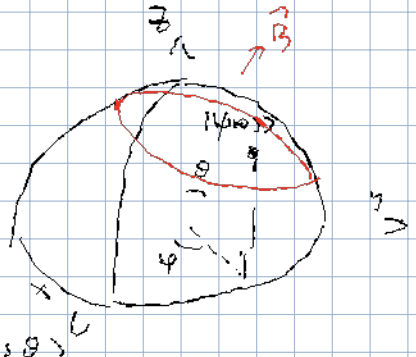
$$= \begin{pmatrix} e^{-iBt/\hbar}\cos\frac{\theta}{2}e^{+i\phi} & e^{iBt/\hbar}\sin\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{+iBt/\hbar}\cos\frac{\theta}{2}e^{-i\phi} \\ e^{-iBt/\hbar}\sin\frac{\theta}{2} \end{pmatrix}$$

$$= \begin{pmatrix} e^{iBt/\hbar}\sin\frac{\theta}{2} & e^{-iBt/\hbar}\cos\frac{\theta}{2}e^{i\phi} \end{pmatrix} \begin{pmatrix} e^{+iBt/\hbar}\cos\frac{\theta}{2}e^{-i\phi} \\ e^{-iBt/\hbar}\sin\frac{\theta}{2} \end{pmatrix}$$

$$= e^{i\frac{2B}{\hbar}t}\sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{-i\phi} + e^{-i\frac{2B}{\hbar}t}\sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{i\phi}$$

$$= \frac{1}{2}\sin\theta\left\{e^{i(\frac{2B}{\hbar}-\phi)t} + e^{-i(\frac{2B}{\hbar}-\phi)t}\right\}$$

$$= \sin\theta\cos(\phi - \omega t) \quad \omega = \frac{2B}{\hbar}$$



$\langle \sigma_y \rangle = \sin\theta\sin(\phi - \omega t)$

$\langle \sigma_z \rangle = \cos\theta$

$\langle \vec{\sigma} \rangle = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$

$= (\sin\theta\cos(\phi-\omega t),\ \sin\theta\sin(\phi-\omega t),\ \cos\theta)$

$\Rightarrow \omega = \frac{2B}{\hbar}$: Larmor 歲差 (Bloch 球上)

5. Heisenberg Picture

$\vec{S}(t=0) = \vec{\sigma}$

Bloch Equation :

$$i\hbar\frac{d\vec{\sigma}}{dt} = [\vec{\sigma}, \vec{H}] = [\vec{\sigma}, -\sigma_z B]$$

$$= -B(-2i\sigma_y, 2i\sigma_x, 0) = 2i\vec{B}\times\vec{\sigma}$$

$\Rightarrow \vec{S}(t) = (\cos\omega t\,\hat{\sigma}_x - \sin\omega t\,\hat{\sigma}_y,\ \sin\omega t\,\hat{\sigma}_x + \cos\omega t\,\hat{\sigma}_y,\ \hat{\sigma}_z)$

$\langle\vec{S}(t)\rangle = \langle\psi(t=0)|\vec{S}(t)|\psi(t=0)\rangle$

$= \begin{pmatrix} \cos\frac{\theta}{2}e^{-i\phi} & \sin\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \cos\omega t\,\hat{\sigma}_x - \sin\omega t\,\hat{\sigma}_y \\ \sin\omega t\,\hat{\sigma}_x + \cos\omega t\,\hat{\sigma}_y \\ \hat{\sigma}_z \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2}e^{-i\phi} \\ \sin\frac{\theta}{2} \end{pmatrix}$

$= \begin{pmatrix} \sin\theta\cos\theta - \omega t & \sin\theta\sin\theta - \omega t \\ \cos\theta & \end{pmatrix}$

7. Time-Dependence

$\vec{B} = \vec{B}(t) = (B_x(t), B_y(t), B_z)$

$H = -\vec{\sigma}\cdot\vec{B} = -B_x(t)\sigma_x - B_y(t)\sigma_y - B_z\sigma_z$

① Eigen-basis.

$i\hbar\partial_t \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix} = H(t) \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}, \quad H_0 = -B_z\sigma_z$

Separate Dynamic Phase:

$$i\hbar\partial_t \left[e^{-i\frac{H_0}{\hbar}t} |\bar{\psi}\rangle \right] = H e^{-i\frac{H_0}{\hbar}t} |\bar{\psi}\rangle$$

$$i\hbar\left[(-i\tfrac{H_0}{\hbar})|\bar{\psi}\rangle + \partial_t|\bar{\psi}\rangle\right] e^{-i\frac{H_0}{\hbar}t} = (H_0 + V) e^{-i\frac{H_0}{\hbar}t} |\bar{\psi}\rangle$$

$$i\hbar\partial_t|\bar{\psi}\rangle = e^{i\frac{H_0}{\hbar}t} V e^{-i\frac{H_0}{\hbar}t} |\bar{\psi}\rangle$$

(Interaction Picture)

$|\Psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$$i\hbar \begin{pmatrix} \dot{\alpha} \\ \dot{\beta} \end{pmatrix} = e^{-i\frac{B_z}{\hbar}\sigma_z t} \begin{pmatrix} 0 & B_x - iB_y \\ B_x + iB_y & 0 \end{pmatrix} e^{i\frac{B_z}{\hbar}\sigma_z t} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Initial Condition: $\begin{pmatrix} \alpha(0) \\ \beta(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\hookrightarrow \dot{\alpha}(0) = 0, \quad \dot{\beta}(0) = e^{-i\frac{B_z}{\hbar}\sigma_z t} \left[B_x(0) + iB_y(0)\right] e^{i\frac{B_z}{\hbar}\sigma_z t}$

If, $B_x(t) = B_x \cos\Omega t, \quad B_y(t) = B_x \sin\Omega t$

$$\hookrightarrow i\hbar \begin{pmatrix} \dot{\alpha} \\ \dot{\beta} \end{pmatrix} = \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} \begin{pmatrix} 0 & B_x e^{-i\Omega t} \\ B_x e^{i\Omega t} & 0 \end{pmatrix} \begin{pmatrix} e^{i\omega t} & 0 \\ 0 & e^{-i\omega t} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

1-st order perturbation.

$$\simeq \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{i\omega t} \end{pmatrix} \begin{pmatrix} 0 & B_x e^{-i\Omega t} \\ B_x e^{i\Omega t} & 0 \end{pmatrix} \begin{pmatrix} e^{i\omega t} & 0 \\ 0 & e^{-i\omega t} \end{pmatrix} \begin{pmatrix} \alpha(0) \\ \beta(0) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & B_x e^{-i(\Omega+\omega)t} \\ B_x e^{i(\Omega+\omega)t} & 0 \end{pmatrix} \begin{pmatrix} e^{i\omega t} & 0 \\ 0 & e^{-i\omega t} \end{pmatrix} \begin{pmatrix} \alpha(0) \\ \beta(0) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & B_x e^{-i(\Omega+2\omega)t} \\ B_x e^{i(\Omega+2\omega)t} & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ B_x e^{i(\Omega+2\omega)t} \end{pmatrix} \quad \omega = \frac{B_z}{\hbar}$$

$$\beta(t) = \beta(0) + \int \frac{1}{i\hbar} B_x e^{i(\Omega+2\omega)t} dt = \frac{B_x}{i\hbar} \frac{e^{i(\Omega+2\omega)t} - 1}{i(\Omega+2\omega)}$$

$$|\beta(t)|^2 = \frac{B_x^2}{\hbar^2(\Omega+2\omega)^2}\left\{\left[\cos(\Omega+2\omega)\frac{t}{2}-1\right]^2+\sin^2(\Omega+2\omega)\frac{t}{2}\right\}$$

$$= \frac{2B_x^2}{\hbar^2(\Omega+2\omega)^2}\left[1-\cos(\Omega+2\omega)\right]$$

$$= \frac{B_x^2}{\hbar^2}\left\{\frac{\sin\left[\frac{1}{2}(\Omega+2\omega)t\right]}{\frac{1}{2}(\Omega+2\omega)}\right\}^2 t$$

$\lim_{t\to\infty}|\beta(t)|^2 = \frac{B_x^2}{\hbar^2}\pi\delta\left(\frac{\Omega}{2}+\omega\right)t$ ← $\lim_{t\to\infty}\frac{\sin^2\omega t}{\omega^2}$ = 적분하면 t.

$$= \frac{2\pi B_x^2}{\hbar}\delta(\hbar\Omega+2B_z)\,t$$

$\int$: Transition Rate

$\Omega = -\frac{2B_z}{\hbar}$ ⇒ $|\psi\rangle : \begin{pmatrix}1\\0\end{pmatrix} \to \begin{pmatrix}0\\1\end{pmatrix}$

$|\beta|^2 \sim P_{|\downarrow\rangle\to|\uparrow\rangle}$

② Transformation :

$\psi(t) = U(t)\,\tilde{\psi}(t)$

$i\hbar\partial_t(U\tilde{\psi}) = H(U\tilde{\psi})$

$i\hbar(\partial_t U)\tilde{\psi} + i\hbar U\partial_t\tilde{\psi} = \tilde{H}U\tilde{\psi}$

$i\hbar\partial_t\tilde{\psi} = U^\dagger\left[\tilde{H}U\tilde{\psi} - i\hbar(\partial_t U)\tilde{\psi}\right]$

$= \left[U^\dagger\tilde{H}U - i\hbar U^\dagger(\partial_t U)\right]\tilde{\psi} = H_{\text{eff}}\,\tilde{\psi}$

Let $U_1 = e^{-i\Omega\frac{t}{2}\sigma_z}$

$U_2 = e^{-i\omega\frac{t}{2}\sigma_z}$

$H_{\text{eff}} = U_1^\dagger H U_1 - i\hbar U_1^\dagger(\partial_t U_1)$

$= e^{i\Omega\frac{t}{2}\sigma_z}\left(-B_x\cos\Omega t\,\sigma_x - B_x\sin\Omega t\,\sigma_y - B_z\sigma_z\right)e^{-i\Omega\frac{t}{2}\sigma_z}$

$- i\hbar\, e^{i\Omega\frac{t}{2}\sigma_z}\left(-\frac{i}{2}\Omega\sigma_z\right)e^{-i\Omega\frac{t}{2}\sigma_z}$

$= -B_x\begin{pmatrix}e^{i\Omega\frac{t}{2}} & 0\\ 0 & e^{-i\Omega\frac{t}{2}}\end{pmatrix}\begin{pmatrix}0 & \cos\Omega t - i\sin\Omega t\\ \cos\Omega t + i\sin\Omega t & 0\end{pmatrix}\begin{pmatrix}e^{-i\Omega\frac{t}{2}} & 0\\ 0 & e^{i\Omega\frac{t}{2}}\end{pmatrix}$

$- e^{i\Omega\frac{t}{2}\sigma_z}\left[B_z + \frac{\hbar}{2}\Omega\right]\sigma_z e^{-i\Omega\frac{t}{2}\sigma_z}$

$= -B_x\sigma_x - \left(B_z + \frac{\hbar}{2}\Omega\right)\sigma_z$

$\vec{B}_{\text{eff}} = (B_x\cos\Omega t, -B_x\sin\Omega t, B_z)$

↓

$\vec{B} = (B_x, 0, B_z + \frac{1}{2}\omega_0)$

$H_{eff_2} = \vec{n_2} \cdot H_{eff_1} \vec{n_2} + \frac{1}{2}\omega_0 \vec{n_2} \cdot (0,0,1)$

$= e^{i\sigma_y \theta_y/2} \left\{ -B_x \sigma_x - (B_z + \frac{1}{2}\omega_0)\sigma_z \right\} e^{-i\sigma_y \theta_y/2}$

其中, $e^{i\sigma_y \theta_y/2} = \sum_k \frac{1}{k!} (\frac{i}{2}\theta_y)^k$

$= \sum_{k=2m} \frac{(-1)^m}{k!} (\frac{\theta}{2})^k \sigma_y^{2m} + i \sum_{k=2m+1} \frac{(-1)^m}{k!} (\frac{\theta}{2})^k \sigma_y^{2m} \cdot \sigma_y \qquad \sigma_y^2 = I$

$= \cos\frac{\theta}{2} I + i \sin\frac{\theta}{2} \sigma_y$

$e^{-i\sigma_y \theta_y/2} = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} \sigma_y$

$H_{eff_2} = \begin{pmatrix} \cos\frac{\theta}{2} & \sin\frac{\theta}{2} \\ -\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} B_z + \frac{1}{2}\omega_0 & B_x \\ B_x & -(B_z + \frac{1}{2}\omega_0) \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$

$= \begin{pmatrix} B_x \sin\frac{\theta}{2} + \cos\frac{\theta}{2}(B_z + \frac{1}{2}\omega_0) & \cos\frac{\theta}{2} B_x - \sin\frac{\theta}{2}(B_z + \frac{1}{2}\omega_0) \\ \cos\frac{\theta}{2}(B_z + \frac{1}{2}\omega_0) - \sin\frac{\theta}{2} B_x & -\sin\frac{\theta}{2}(B_z + \frac{1}{2}\omega_0) - \cos\frac{\theta}{2} B_x \end{pmatrix} \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$

$B_z' = B_z + \frac{1}{2}\omega_0$

$= \begin{pmatrix} B_z'(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) + 2B_x \sin\frac{\theta}{2}\cos\frac{\theta}{2} & -B_z' 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} + B_x(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) \\ -B_z' 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} + B_x(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) & -B_z'(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) - 2B_x \sin\frac{\theta}{2}\cos\frac{\theta}{2} \end{pmatrix}$

$= (B_z' \cos\theta + B_x \sin\theta)\sigma_z - (B_z' \sin\theta - B_x \cos\theta)\sigma_y$

$= |B_z'^2 + B_x^2|^{\frac{1}{2}} \left[\sin(\theta - \varphi)\sigma_z - i\cos(\theta - \varphi)\sigma_y \right]$

其中 $\varphi = \arctan(\frac{B_z'}{B_x})$

取 $\eta = \frac{1}{2}(\theta - \varphi)$ $\Rightarrow H_{eff_2} = |B_z'^2 + B_x^2|^{\frac{1}{2}} \sigma_z$

$|\psi(t)\rangle \longleftarrow \vec{n} \longleftarrow |\vec{n}\rangle \longleftarrow e^{-i\sigma\cdot\vec{n}\varphi/2}|\vec{n}\rangle$

$|\vec{n}\rangle = |\vec{n}\rangle$

$\langle\vec{n}|\vec{n}\rangle = \langle\vec{n}|\vec{n}\rangle$

$|\vec{\Omega}(t)\rangle = e^{-i\Omega t H_{eff_2}/2}|\vec{\Omega}(0)\rangle$

$|\psi(t)\rangle = U(t,0)|\psi(0)\rangle$

$= U_1(t) U_2 e^{-i H_{eff_2}(t-0)/2} U_2^\dagger U_1^\dagger |\psi(0)\rangle$

$= e^{-i\sigma_z \omega_0 t/2} e^{-i\sigma_y \theta_y/2} e^{i\sigma_z \theta_z/2} e^{i\sigma_y \theta_y/2} e^{i\sigma_z \omega_0 t/2} |\psi(0)\rangle$

$$= \begin{pmatrix} e^{-i\Omega\Delta t/2} & 0 \\ 0 & e^{i\Omega\Delta t/2} \end{pmatrix} \begin{pmatrix} \cos\eta/2 & -\sin\eta/2 \\ \sin\eta/2 & \cos\eta/2 \end{pmatrix} \begin{pmatrix} e^{i\omega_{bare}\Delta t/2} & 0 \\ 0 & e^{-i\omega_{bare}\Delta t/2} \end{pmatrix}$$

$$\begin{pmatrix} \cos\eta/2 & \sin\eta/2 \\ -\sin\eta/2 & \cos\eta/2 \end{pmatrix} \begin{pmatrix} e^{i\Omega\Delta t/2} & 0 \\ 0 & e^{-i\Omega\Delta t/2} \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\Omega\Delta t/2}\cos\eta/2 & -e^{-i\Omega\Delta t/2}\sin\eta/2 \\ e^{i\Omega\Delta t/2}\sin\eta/2 & e^{i\Omega\Delta t/2}\cos\eta/2 \end{pmatrix} \begin{pmatrix} e^{i\omega_{bare}\Delta t/2} & 0 \\ 0 & e^{-i\omega_{bare}\Delta t/2} \end{pmatrix}$$

$$\begin{pmatrix} e^{i\Omega\Delta t/2}\cos\eta/2 & e^{-i\Omega\Delta t/2}\sin\eta/2 \\ -e^{i\Omega\Delta t/2}\sin\eta/2 & e^{-i\Omega\Delta t/2}\cos\eta/2 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i(\frac{\Omega}{2}-\frac{\omega_{bare}}{2})\Delta t}\cos\eta/2 & -e^{-i(\frac{\Omega}{2}+\frac{\omega_{bare}}{2})\Delta t}\sin\eta/2 \\ e^{i(\frac{\Omega}{2}+\frac{\omega_{bare}}{2})\Delta t}\sin\eta/2 & e^{i(\frac{\Omega}{2}-\frac{\omega_{bare}}{2})\Delta t}\cos\eta/2 \end{pmatrix} \begin{pmatrix} e^{i\Omega\Delta t/2}\cos\eta/2 & e^{-i\Omega\Delta t/2}\sin\eta/2 \\ -e^{i\Omega\Delta t/2}\sin\eta/2 & e^{-i\Omega\Delta t/2}\cos\eta/2 \end{pmatrix}$$

$\omega = \frac{B_{bare}}{\hbar}$

$$= \begin{pmatrix} e^{-i\Omega\Delta t/2}\left[e^{i\omega\Delta t}\cos^2\frac{\eta}{2} + e^{-i\omega\Delta t}\sin^2\frac{\eta}{2}\right] & e^{-i\Omega\Delta t/2}\left(e^{i\omega\Delta t}-e^{-i\omega\Delta t}\right)\sin\frac{\eta}{2}\cos\frac{\eta}{2} \\ e^{i\Omega\Delta t/2}\left(e^{i\omega\Delta t}-e^{-i\omega\Delta t}\right)\sin\frac{\eta}{2}\cos\frac{\eta}{2} & e^{i\Omega\Delta t/2}\left[e^{i\omega\Delta t}\sin^2\frac{\eta}{2} + e^{-i\omega\Delta t}\cos^2\frac{\eta}{2}\right] \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\Omega\Delta t/2}\left[\cos(\omega\Delta t) + i\sin(\omega\Delta t)\cos\eta\right] & i e^{-i\Omega\Delta t/2}\sin(\omega\Delta t)\sin\eta \\ i e^{i\Omega\Delta t/2}\sin(\omega\Delta t)\sin\eta & e^{i\Omega\Delta t/2}\left[\cos(\omega\Delta t) - i\sin(\omega\Delta t)\cos\eta\right] \end{pmatrix}$$

$t = 0$: $\vec{B} = (B_x, 0, B_z)$

eigenstate : $\psi(0) = \begin{pmatrix} \cos\frac{\alpha}{2} \\ \sin\frac{\alpha}{2} \end{pmatrix}$, $\tan\alpha = \frac{B_x}{B_z}$

$$\psi(t) = \begin{pmatrix} e^{-i\Omega t/2}\left\{\cos\omega t\cos\frac{\alpha}{2} + i\sin\omega t\left[\cos\eta\cos\frac{\alpha}{2} + \sin\eta\sin\frac{\alpha}{2}\right]\right\} \\ e^{i\Omega t/2}\left\{\cos\omega t\sin\frac{\alpha}{2} + i\sin\omega t\left[\sin\eta\cos\frac{\alpha}{2} - \cos\eta\sin\frac{\alpha}{2}\right]\right\} \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\Omega t/2}\left[\cos\omega t\cos\frac{\alpha}{2} + i\sin\omega t\cos(\eta-\frac{\alpha}{2})\right] \\ e^{i\Omega t/2}\left[\cos\omega t\sin\frac{\alpha}{2} + i\sin\omega t\sin(\eta-\frac{\alpha}{2})\right] \end{pmatrix}$$

If : $\Omega \to 0$, $\eta \simeq \alpha$

$$\omega = \frac{1}{\hbar}\sqrt{(B_z - \frac{1}{2}\hbar\Omega)^2 + B_x^2} \simeq \frac{1}{\hbar}\sqrt{B_z^2 + B_x^2} - \frac{1}{2}\frac{\hbar\Omega B_z}{\sqrt{B_z^2+B_x^2}}$$

$$= \frac{1}{\hbar}\left(\sqrt{B_z^2 + B_x^2} - \frac{1}{2}\hbar\Omega\cos\alpha\right)$$

Adiabatic

$$\begin{pmatrix} \cos\frac{\alpha}{2}e^{i(\omega-\frac{\Omega}{2})t} & -\sin\frac{\alpha}{2}e^{-i(\omega+\frac{\Omega}{2})t} \\ \sin\frac{\alpha}{2}e^{i(\omega+\frac{\Omega}{2})t} & \cos\frac{\alpha}{2}e^{-i(\omega-\frac{\Omega}{2})t} \end{pmatrix} \begin{pmatrix} e^{-i(\cos\alpha+1)\phi} & 0 \\ 0 & e^{i(\cos\alpha+1)\phi} \end{pmatrix} e^{-i\sqrt{B_z^2+B_x^2}\,t}$$

$\left(T_\phi = \frac{2\pi}{\omega}\right)$ Geometric Phase. Dynamic Phase

$|\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$|\psi_1(t)\rangle = U_{21}(t,0)\,|\psi_1(0)\rangle = -i e^{\frac{i\Omega t}{2}} \sin(\omega t)\sin\eta$

$|\psi_1(t)|^2 = \sin^2\omega t \sin^2\eta$

$= \sin^2\left[\frac{1}{4}\left(B_z + \frac{\hbar\Omega}{2}\right)^2 + B_x^2\right] \frac{B_x^2}{\left(B_z + \frac{\hbar\Omega}{2}\right)^2 + B_x^2}$

$\Omega = -\frac{2B_z}{\hbar}$, $|\psi_1(t)|^2 \to 1$

Rabi Resonance

Linear Oscillation:

$\vec{B} = (B_x \cos\Omega t,\ 0,\ B_z)$

$U_1 = e^{-\frac{i\Omega t\sigma_z}{2}}$

$H_{eff} = U_1^\dagger H U_1 - i\hbar U_1^\dagger(\partial_t U_1)$

$= e^{\frac{i\Omega t\sigma_z}{2}}\left(-B_x\cos\Omega t\,\sigma_x - B_z\sigma_z\right)e^{-\frac{i\Omega t\sigma_z}{2}} - \frac{1}{2}\hbar\Omega\sigma_z$

$= -B_x\cos\Omega t \begin{pmatrix} e^{\frac{i\Omega t}{2}} & 0 \\ 0 & e^{-\frac{i\Omega t}{2}} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} e^{-\frac{i\Omega t}{2}} & 0 \\ 0 & e^{\frac{i\Omega t}{2}} \end{pmatrix}$

$- \left(B_z + \frac{1}{2}\hbar\Omega\right)\sigma_z$

$= -B_x\cos\Omega t \begin{pmatrix} 0 & e^{i\Omega t} \\ e^{-i\Omega t} & 0 \end{pmatrix} - \left(B_z + \frac{1}{2}\hbar\Omega\right)\sigma_z$

$= -B_x\cos\Omega t\,(\cos\Omega t\,\sigma_x - \sin\Omega t\,\sigma_y) - \left(B_z + \frac{1}{2}\hbar\Omega\right)\sigma_z$

$\approx -\frac{1}{2}B_x\sigma_x - \left(B_z + \frac{1}{2}\hbar\Omega\right)\sigma_z$ Drop 2Ω term.

↝ Rotational Oscillation.

Ex Ramsey Interferometry.

$H = -B_z\sigma_z - B_x\sin\Omega t\cos\Omega t\,\sigma_x = \begin{cases} -B_z\sigma_z - B_x\cos\Omega t\,\sigma_x \\ -B_z\sigma_z \end{cases}$

$s(t) = \begin{cases} 1 & 0 < t < \tau \\ 0 & \tau < t < \tau + T \\ 1 & \tau + T < t < 2\tau + T \end{cases}$

$U(2\tau+T) = U_1(T+2\tau, T+\tau)\,U_0(T+\tau, \tau)\,U(\tau,0)\,\psi(0)$

$$\frac{q}{2\hbar B_z} \to \theta \;,\quad 0 \leftarrow \varphi \quad B_x \to 0$$

$$\left|\psi(\theta)\right\rangle = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}$$

$$(L_z+L_-)\psi_\uparrow = -2\hbar\sin\theta\sin\varphi\,(\cos\theta\cos\phi - \sin\theta\cos\varphi\sin\phi)$$

$$|(L_z+L_-)\psi_\uparrow|^2 = 4\hbar^2\sin^2\theta\sin^2\varphi\,(\cos\theta\cos\phi - \sin\theta\cos\varphi\sin\phi)^2$$

$$\psi = \frac{B_z q}{L_z}$$

$$(L_z+L_-)\psi$$

$$= \begin{pmatrix} \lambda_+\cos\theta\sin i - \theta\cos & \lambda_+\sin\theta\sin\varphi \\ -\lambda_+\sin\theta\sin\varphi & \cos\theta - i\sin\theta\cos\varphi \end{pmatrix} \begin{pmatrix} e^{i\phi} & \\ & e^{-i\phi} \end{pmatrix} \begin{pmatrix} \cos\theta + i\sin\theta\cos\varphi & i\lambda_+\sin\theta\sin\varphi \\ -i\lambda_+\sin\theta\sin\varphi & \cos\theta - i\sin\theta\cos\varphi \end{pmatrix} =$$

$$= \begin{pmatrix} (\cos\theta - i\sin\theta\cos\varphi)\,e^{i\phi} & i\lambda_+\sin\theta\sin\varphi\; e^{-i\phi} \\ -i\lambda_+\sin\theta\sin\varphi\; e^{i\phi} & (\cos\theta + i\sin\theta\cos\varphi)\,e^{-i\phi} \end{pmatrix} \begin{pmatrix} \cos\theta + i\sin\theta\cos\varphi & i\lambda_+\sin\theta\sin\varphi \\ -i\lambda_+\sin\theta\sin\varphi & \cos\theta - i\sin\theta\cos\varphi \end{pmatrix} =$$

$$= \begin{pmatrix} (\cos\theta - i\sin\theta\cos\varphi)\,e^{i\phi} + \lambda_+^2\sin^2\theta\sin^2\varphi\; e^{-i\phi} & i\lambda_+\sin\theta\sin\varphi\,(e^{i\phi}(\cos\theta + i\sin\theta\cos\varphi) + e^{-i\phi}(\cos\theta - i\sin\theta\cos\varphi)) \\ -i\lambda_+\sin\theta\sin\varphi\,(e^{i\phi}(\cos\theta + i\sin\theta\cos\varphi) + e^{-i\phi}(\cos\theta - i\sin\theta\cos\varphi)) & (\cos\theta + i\sin\theta\cos\varphi)^2 e^{i\phi} + \lambda_+^2\sin^2\theta\sin^2\varphi\; e^{-i\phi} \end{pmatrix} =$$

$$= \begin{pmatrix} e^{-i\phi}(\cos\theta - i\sin\theta\cos\varphi)(e^{i\phi}(\cos\theta - i\sin\theta\cos\varphi) + \lambda_+^2\sin^2\theta\sin^2\varphi\; e^{-i\phi}) & i\lambda_+\sin\theta\sin\varphi\,(e^{i\phi}(\cos\theta - i\sin\theta\cos\varphi) + e^{-i\phi}(\cos\theta + i\sin\theta\cos\varphi)) \\ -i\lambda_+\sin\theta\sin\varphi\,(e^{i\phi}(\cos\theta - i\sin\theta\cos\varphi) + e^{-i\phi}(\cos\theta + i\sin\theta\cos\varphi)) & e^{i\phi}(\cos\theta + i\sin\theta\cos\varphi)^2 + \lambda_+^2\sin^2\theta\sin^2\varphi\; e^{-i\phi} \end{pmatrix} =$$

$$= \begin{pmatrix} \lambda_+\cos\theta\sin i - \theta\cos & \lambda_+\sin\theta\sin\varphi \\ -\lambda_+\sin\theta\sin\varphi & \cos\theta - i\sin\theta\cos\varphi \end{pmatrix} \begin{pmatrix} \cos\theta + i\sin\theta\cos\varphi & i\lambda_+\sin\theta\sin\varphi \\ -i\lambda_+\sin\theta\sin\varphi & \cos\theta - i\sin\theta\cos\varphi \end{pmatrix} =$$

$$= \cos\theta\, I + i\sin\theta\,(\lambda_+\sin\varphi\,\sigma_x + \cos\varphi\,\sigma_y) =$$

$$B_{tot} = \sqrt{B_z^2 + B_x^2}$$

$$\theta = \frac{B_{tot}}{\hbar}\tau$$

$$\varphi = \arctan\left(\frac{B_x}{B_z}\right)$$

$$= \sum_{k=2m} (-1)^m \frac{\lambda^k}{k!\,\hbar^k} B_{tot}^k + i\sum_{k=2m+1} (-1)^m \frac{\lambda^k}{k!\,\hbar^k} B_{tot}^k\,(B_z'\sigma_z + B_x\sigma_x) =$$

$$B_{tot} = B_{tot}(B_z', B_x)$$

$$= \sum_{k=2m} (-1)^m \frac{\lambda^k}{k!\,\hbar^k} B_{tot}^k + i\sum_{k=2m+1} (-1)^m \frac{\lambda^k}{k!\,\hbar^k} B_{tot}^k\,(B_z'\sigma_z + B_x\sigma_x) = \sum_{k=1}^{\infty} \frac{\lambda^k}{k!\,\hbar^k} (B_z'\sigma_z + q_x\sigma_x)^k = e^{\frac{q}{i\hbar}\int_0^\tau (B_z'\sigma_z + B_x\sigma_x)\,d\tau} = e^{\frac{q}{i\hbar}\int_0^\tau (B_z'\sigma_z + B_x\sigma_x)\,d\tau}$$

$$= e^{\frac{q}{i\hbar}\int_0^\tau (B_z'\sigma_z + B_x\sigma_x)\,d\tau} = e^{\frac{q}{i\hbar}\int_0^\tau (B_z'\sigma_z + B_x\sigma_x)\,d\tau} \qquad \psi(0)$$

$$|\psi_+(\tau+\tau_2)|^2 \simeq \varphi \sin^2\left(\frac{B_z \tau_2}{\hbar}\right)\left(\frac{B_x}{B_0}\right)^2 \left[\cos\theta\cos\phi - \sin\vartheta \sin\phi\right]^2 \qquad \omega = \frac{B_0}{\hbar}$$

$$= \varphi \sin^2(\omega\tau)\left(\frac{B_x}{\hbar\omega}\right)^2 \cos^2\left[\omega(\tau+\tau)\right]$$